

Application Programming Interface

- It is a way where client-server can communicate basis on request with each other
- Common API standard used by most mobile/web applications to communicate with servers is called **REST**
- REST stands for REpresentational State Transfer.
- It is new set of rules that has been the common standard for building web API.
- An API which follows REST standard is called RESTful API
- Examples of RESTful API -> twilio, stripe, Google Maps

Basics of REST API

- A RESTful API will organizes resources into a set of unique URIs(Unique Resource Identifiers)
- Example -> <https://example.com/api/v3/products> or <https://example.com/api/v3/users>
- URIs differentiate different types of resources on a server.
- The **Resources** should be group by *noun* and not *verbs*. Good: /users -> Bad: /createUsers

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- A client interacts with a resource by making a request to the endpoint for the resource over HTTP.
 - HTTP **methods** (GET, POST, PUT, DELETE) define the action. Bad: POST /createUsers -> Good: POST /users.
 - POST -> Create a resource, GET -> Read the data on resource.
 - PUT -> Updating an existing resource, DELETE -> Delete a resource. **CRUD Operations**

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- HTTP status code. 200 means Success, 400 means Something wrong with our request, 500 means Something wrong at server level
 - POST is **not idempotent**, while other methods are. An operation is **idempotent** if making the same request multiple times has the same effect as making it once.
 - POST is used to create new resources or trigger actions. Other methods are designed to be idempotent.
 - Response body is optional, could contain data payload and usually formatted in JSON -> {'key':'value'} -> not dictionary

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- REST implementation should be stateless, meaning the two parties don't need to store any information about each other and every request and response(cycle) is independent from all others.
 - It leads to web applications that are easy to scale and well behaved.
 - Two Final Points

1. If an API endpoints returns huge amount of data, then use *pagination*. A common pagination scheme uses limit & offset as parameters.

Example -> `/products?limit=25&offset=50` . If they are not specified, the server should assume sensible default values.

2. Versioning of an API is very important. It allows an implementation to provide backward compatibility, so, that if we introduce breaking changes from one version to another, consumers get enough time to move to the next version. The most straightforward way to version an API is to prefix the version before the resource on the URI. Example -> `/v1/products` --> `/v2/products`

RESTful API is simple and effective when applied sensibly. It may not be the best choice for all companies, but still it is used widely all around the globe.

Other popular APIs option is **GraphQL** and **gRPC**.

Resource: <https://www.youtube.com/watch?v=-mN3VyJuCjM>